$$\langle x|F(p)|x'\rangle = \sum_j F_j \left(\frac{\hbar}{i}\frac{d}{dx}\right)^j \delta(x-x')$$

$$= F\left(\frac{\hbar}{i}\frac{d}{dx}\right)\delta(x-x') \quad *$$

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3. Consider a quantum system with quantum state $|4\rangle$ which is translated by a distance ξ, i.e.

$$|4'\rangle = U(P;\xi)|4\rangle = e^{-\frac{i}{\hbar}\xi P}|4\rangle$$

Find the q-representation of $|4'\rangle$ explicitly and interpret your result.

$$U(P;\xi)|x\rangle = |x+\xi\rangle$$

$$\langle x|\psi'\rangle = \langle x|U|4\rangle = \int_{-\infty}^{\infty}\langle x|U|x'\rangle\langle x'|4\rangle\,dx'$$

$$= \int_{-\infty}^{\infty}\underbrace{\langle x|x'+\xi\rangle}_{\delta(x-x'-\xi)}\langle x'|4\rangle\,dx'$$

$$= \langle x-\xi|4\rangle$$

$$= \psi(x-\xi)$$

The effect of translation operator on position basis is to change $|x\rangle$ to $|x+\xi\rangle$, which is equivalent to changing the q-representation of ψ from $\psi(x)$ to $\psi(x-\xi)$. The same property holds in classical mechanics considering transformation.